# Andrea Churchwell

Selmer, TN
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## Education

**Columbia University** **Aug 2024 – 2026 (Expected)**
*Applied AI Solutions Engineering* *Remote*

**Persevere Tech Alliance** **2023 – 2024**
*Full-Stack Web Development Bootcamp (MERN Stack)* *Remote*

## Professional Experience

**Dental Assistant** **2002 – 2017**
*Private Dental Practice* *Selmer, TN*

## Technical Skills

**Programming:** Python, JavaScript, SQL, HTML/CSS
**Full-Stack Development:** React, FastAPI, SQLite, Git, GitHub
**Data & Machine Learning:** Pandas, NumPy, Scikit-learn, Plotly

## Projects

**Our Door – AI-Powered Learning Platform** **2026**
*React, FastAPI, SQLite, ChromaDB, OpenAI, JWT Authentication*

- Built a full-stack web application designed to help students learn through guided problem solving rather than direct answers.
- Developed a React frontend and FastAPI backend supporting both student and administrator workflows.
- Implemented JWT-based authentication and role-based access control for secure access management.
- Integrated retrieval-augmented generation (RAG) using ChromaDB and OpenAI embeddings to provide curriculum-aware responses.
- Created instructor analytics and logging features to surface common student questions and learning challenges.
- Collaborated on a three-person development team using Git, GitHub, pull requests, and Agile-style standups.

**Open Source Contributor – CesiumJS** **2026 – Present**
*JavaScript, Git, GitHub, Open Source Development*

- Contributing to the CesiumJS open-source codebase through the Fair Chance Futures program.
- Investigating reported issues and navigating a large production codebase using Git, pull requests, and issue tracking workflows.

**Andrea's Bracket Breakdown – March Madness Simulator | Live Demo** **2026**
*Python, Pandas, Scikit-learn, NumPy, Streamlit, Plotly*

- Developed a machine learning-powered March Madness bracket simulator using historical NCAA tournament data.
- Engineered team-level and matchup-level features including scoring margin, win/loss differentials, and scoring statistics.
- Trained a logistic regression model on historical tournament data to predict game outcomes.
- Implemented Monte Carlo simulation (1000+ runs) to model tournament uncertainty and generate team advancement probabilities.
- Developed an interactive Streamlit application to visualize deterministic and probabilistic bracket predictions.
- Deployed the application on Streamlit Community Cloud for real-time user interaction.

## Leadership / Involvement

**Rise Up Conference – Fair Chance Futures** **2025**
*Guest Panel Speaker* *Virtual*

- Invited to speak on a panel discussing technology, AI, and the importance of diverse perspectives in software development.